

Entropy

Clausius inequality

$$\oint \frac{\delta Q}{T} \leq 0$$

$$\begin{cases} \text{Reversible: } \oint \frac{\delta Q}{T} = 0 \\ \text{Irreversible: } \oint \frac{\delta Q}{T} < 0 \end{cases}$$

Proof:

Consider an engine working between two constant- T reservoirs:

$$\begin{cases} \epsilon = 1 - \frac{Q_L}{Q_H} \\ \epsilon \leq \epsilon_{\text{Carnot}} = 1 - \frac{T_L}{T_H} \end{cases}$$

$$\Rightarrow \frac{Q_L}{Q_H} \geq \frac{T_L}{T_H}$$

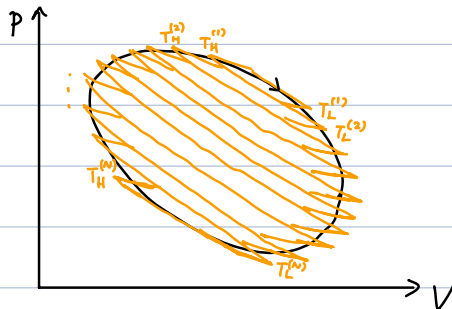
Heat absorbed Heat released

$$\Rightarrow \frac{Q_H}{T_H} - \frac{Q_L}{T_L} \leq 0$$

Define ΔQ to be heat absorbed

$$\Rightarrow \frac{\Delta Q_H}{T_H} + \frac{\Delta Q_L}{T_L} \leq 0$$

For an arbitrary quasi-static cyclic process (more than 2 reservoirs)



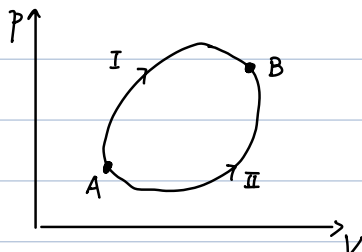
$$\frac{\Delta Q_H^{(i)}}{T_H^{(i)}} + \frac{\Delta Q_L^{(i)}}{T_L^{(i)}} \leq 0 \quad i=1, 2, \dots$$

$$\Rightarrow \sum_i \left(\frac{\Delta Q_H^{(i)}}{T_H^{(i)}} + \frac{\Delta Q_L^{(i)}}{T_L^{(i)}} \right) \leq 0$$

$$\Rightarrow \oint \frac{\delta Q}{T} \leq 0 \quad \text{proved.}$$

Quasi-static reversible cycle:

$$\oint \frac{\delta Q}{T} = 0 \quad (\text{Clausius equality})$$



$$\int_I \frac{\delta Q}{T} = \int_{II} \frac{\delta Q}{T}$$

Reminder: Conservative force

$$\oint \vec{F} \cdot d\vec{r} = 0 \quad \text{or} \quad \int_I \vec{F} \cdot d\vec{r} = \int_{II} \vec{F} \cdot d\vec{r}$$
$$\Leftrightarrow \Delta U = - \int_{r_i}^{r_f} \vec{F} \cdot d\vec{r}$$

Potential energy does NOT depend on path!

Entropy: a state function that is independent of path

$$S_B - S_A = \int_A^B \frac{\delta Q}{T} \quad \delta Q: \text{heat absorbed}$$

Differential form:

$$dS = \frac{\delta Q}{T}$$

reminder: inexact vs exact differential

$$\begin{cases} \delta Q : \text{inexact} \\ \frac{\delta Q}{T} : \text{exact} \end{cases}$$

Entropy calculation

Say. S_A is known

$$\Rightarrow S_B = S_A + \int_A^B \frac{\delta Q}{T} \quad (\text{reminder: energy})$$

Entropy of ideal gas

$$dU = \delta Q - p dV$$

$$= T dS - p dV$$

$$\Rightarrow dS = \frac{1}{T} (dU + p dV)$$

$$pV = nRT \Rightarrow \frac{p}{T} = \frac{nR}{V}$$

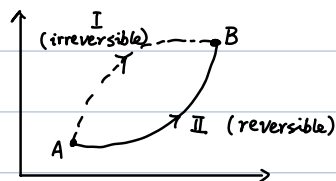
$$= nC_v \frac{dT}{T} + nR \frac{dV}{V}$$

$$\Rightarrow S = S_0 + nC_v \ln \frac{T}{T_0} + nR \ln \frac{V}{V_0}$$

Back to Clausius inequality:

$$\oint \frac{\delta Q}{T} < 0 \quad \text{for irreversible process}$$

Example.



One complete cycle:

$A \xrightarrow{I} B \xrightarrow{II} A$: irreversible

$$\int_{A(I)}^B \frac{\delta Q}{T} + \int_{B(II)}^A \frac{\delta Q}{T} < 0$$

$$\Rightarrow \int_{A(I)}^B \frac{\delta Q}{T} < \int_{A(II)}^B \frac{\delta Q}{T}$$

$$\Rightarrow S_B - S_A > \int_{A(I)}^B \frac{\delta Q}{T}$$

$$\Rightarrow \boxed{S_B - S_A \geq \int_A^B \frac{\delta Q}{T}} \quad \text{Any path from } A \rightarrow B$$

Adiabatic process:

$$\delta Q = 0$$

$$\Rightarrow S_B - S_A \geq 0 \quad (\text{adiabatic})$$

Isolated system (no matter/energy exchange with surrounding)

$$\Rightarrow \text{no heat in/out} \quad (\text{adiabatic})$$

The second law of thermodynamics:

$$\boxed{\Delta S \geq 0} \quad \text{for isolated system}$$

Note:

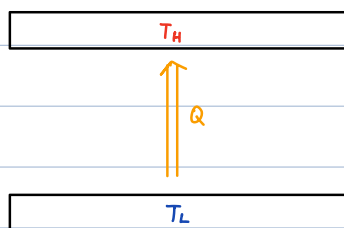
For part of an isolated system (NOT isolated), ΔS can decrease.

For isolated system,

$$\begin{cases} \Delta S = 0 & \Leftrightarrow \text{reversible} \\ \Delta S > 0 & \Leftrightarrow \text{irreversible} \end{cases}$$

$$\Rightarrow \Delta S_{\text{universe}} > 0$$

check Clausius statement

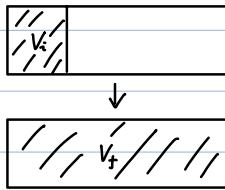


Heat Q from T_L to T_H without other effect
(violation of Clausius)

$$\Rightarrow \Delta S = \frac{Q}{T_H} - \frac{Q}{T_L} < 0$$

(violation of $\Delta S \geq 0$)

Example. Adiabatic free expansion



Not quasi-static !

However, initial & final states are equilibrium states.

$$\Delta Q = 0, \quad W = 0$$

$$U_i = U_f \Rightarrow T_i = T_f$$

$$\begin{aligned} \Delta S &= S_o + n c_v \ln \frac{T_f}{T_o} + n R \ln \frac{V_f}{V_o} \\ &\quad - \left(S_o + n c_v \ln \frac{T_i}{T_o} + n R \ln \frac{V_i}{V_o} \right) \\ &= n R \ln \frac{V_f}{V_i} > 0 \quad \text{irreversible !} \end{aligned}$$

Example. entropy of water at T

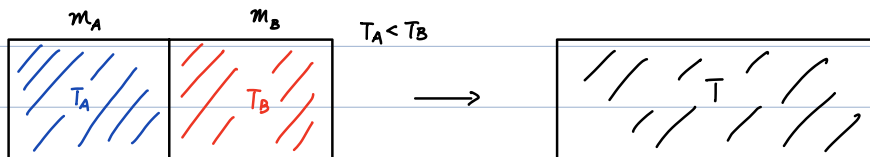
Imagine a quasi-static process:

$$dQ = m c_v dT$$

$$\Rightarrow dS = m c_v \frac{dT}{T}$$

$$\Rightarrow S = m c_v \ln T + \text{constant}$$

Example. mixing water



$$T_A \rightarrow T: \quad \Delta Q_A = m_A c (T - T_A)$$

$$T_B \rightarrow T: \quad \Delta Q_B = m_B c (T - T_B)$$

$$\Delta Q_A + \Delta Q_B = 0 \quad (\text{The full system is adiabatic})$$

$$\Rightarrow T = \frac{m_A T_A + m_B T_B}{m_A + m_B}$$

Again, { NOT quasi-static ;

However, initial & final states are equilibrium states.

$$\begin{cases} \Delta S_A = m_A c_v \ln \frac{T}{T_A} \\ \Delta S_B = m_B c_v \ln \frac{T}{T_B} \end{cases}$$

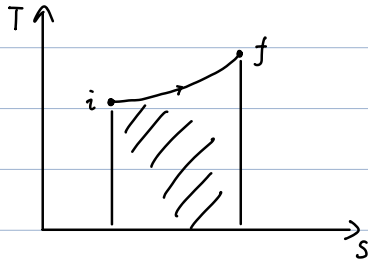
$$\Rightarrow \Delta S = \Delta S_A + \Delta S_B$$

$$= m_A c_v \ln \frac{m_A + m_B T_B / T_A}{m_A + m_B} + m_B c_v \ln \frac{m_A T_A / T_B + m_B}{m_A + m_B}$$

Take $m_A = m_B$ for example.

$$\begin{aligned}\Delta S &= m C_v \ln \frac{m + m T_B/T_A}{2m} + m C_v \ln \frac{m T_A/T_B + m}{2m} \\ &= m C_v \ln \frac{(T_A + T_B)^2}{4 T_A T_B} \\ &> 0 \quad \text{irreversible} \end{aligned}$$

T-S diagram



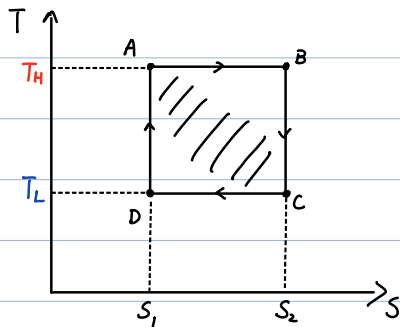
$$dS = \frac{\delta Q}{T}$$

$$\Rightarrow \delta Q = T dS$$

$$\Rightarrow Q = \int_i^f T dS$$

(area in T-S diagram)

Example. Carnot cycle.



$$\begin{cases} B \rightarrow C & \text{adiabatic} \Rightarrow Q=0 \\ D \rightarrow A & \Rightarrow S \text{ is constant} \end{cases}$$

$$W = Q = \oint T dS$$

$$= (T_H - T_L) (S_2 - S_1)$$

$$Q_H = T_H (S_2 - S_1)$$

$$\Rightarrow \epsilon_{\text{Carnot}} = \frac{W}{Q_H} = 1 - \frac{T_L}{T_H}$$